

Four-Bar EV Charging Mechanism Report

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Abstract—Electric vehicle charging automation requires precise, repeatable mechanical systems. This project presents the design, analysis, and prototyping of an autonomous four-bar mechanism to extend an EV-charging end effector from inside a stationary frame to a vehicle charging port, enabling automated and reliable EV charging. The goal was to produce a mechanically simple, repeatable motion that stays within a 400 mm x 450 mm x 500 mm frame when retracted, exits a slit no larger than 101.6 mm x 101.6 mm, and reaches the charging port center located 152.4 mm from the slit at a height of 240 mm. Kinematic synthesis and constraint optimization using MATLAB determined link geometry and input angles. The mechanism was driven by a single HS-425BB servo and controlled by an Arduino receiving an IR trigger (TSOP38238, command 0x10). Simulation trajectory error was up to 0.05 mm; the physical prototype (constructed from 1/4" MDF) reached the target within approximately 2 mm. IR-trigger reliability was improved with protocol filtering and a 500 ms debounce. A torque vs. input-angle analysis indicates the HS-425BB (spec. $\approx 4.8 \text{ kg} \cdot \text{cm} \approx 470 \text{ N} \cdot \text{mm}$) supplies sufficient torque for the motion. The system satisfied the project functional requirements and demonstrates robust integration of mechanical synthesis, control, and sensing for autonomous EV charging. MATLAB and CAD simulations indicate that the design would achieve comparable performance if prototyped.

I. INTRODUCTION

A. Motivation and Context

Electric vehicle (EV) charging infrastructure currently relies on manual interaction in many settings. Automating the physical connection to EV charging ports reduces human effort and exposure to hazards, while improving accessibility and repeatability of charging operations. This work demonstrates a compact, reliable mechanical solution that performs a point-to-point reach inside a constrained frame, minimizing software complexity by leveraging a simple four-bar mechanism.

B. Project Objectives and Functional Requirements

The mechanism was designed to meet the MMAE 232 functional specifications:

- Four-bar linkage fits completely inside a 400 mm x 450 mm x 500 mm foam-core frame when retracted.
- End effector must extend through a slit no larger than 101.6 mm x 101.6 mm.
- End effector must reach the charging port center located 152.4 mm from the slit at a height of 240 mm.
- Single HS-425BB servo provides the actuation; Arduino provides control.

- External IR TSOP38238 receiver triggers motion on command 0x10.
- Motion must be smooth and gentle (no deflection of EV), and hold start and end positions for $\geq 3 \text{ s}$.
- Repeatable operation and LED visual indicator during "charging".

Figure 1 shows the prototype evaluated prior to inserting it into the frame for testing.

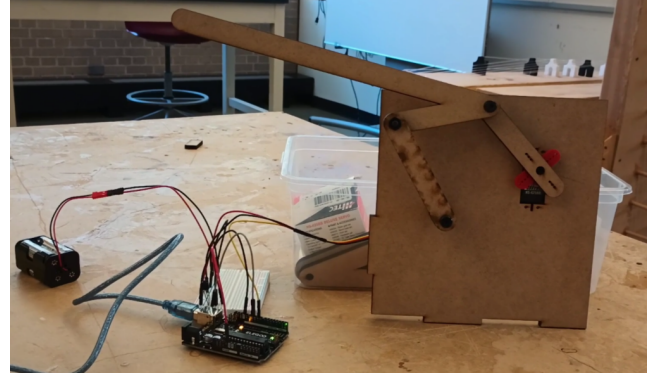


Fig. 1: Prototype used for testing.

II. CONCEPT GENERATION AND EVALUATION

A. Initial Sketches and Precision Points

The four-bar mechanism geometry was initially defined by assigning arbitrary link lengths in MATLAB. These initial lengths serve as a starting point for MATLAB optimization to minimize end-effector positional error while satisfying frame and slit constraints. The design goal was to move the end effector from a fully retracted position inside the frame to the precision point located 152.4 mm from the slit at a height of 240 mm. The chosen parameterization (inputs to the MATLAB solver) used the following reference geometry (all values in mm):

$$x_1 = 280.5 \text{ (pivot x location)}$$

$$y_1 = 132 \text{ (pivot y location)}$$

$$\gamma = -27.2^\circ, \delta = 25.5^\circ$$

The initial link lengths (pre-optimization) were:

$$L_{\text{crank}} = 68 \text{ mm}, \quad L_{\text{coupler}} = 75 \text{ mm}, \quad (1)$$

$$L_{\text{follower}} = 85 \text{ mm}, \quad L_{\text{ground}} = 87 \text{ mm}. \quad (2)$$

with an end-effector reach parameter $r_{ee} = 249.3$ mm.

1) *Three-Point Circle Sketches*: Figure 2 and Figure 3 show hand-drawn three-point circle sketches illustrating the geometric approach.

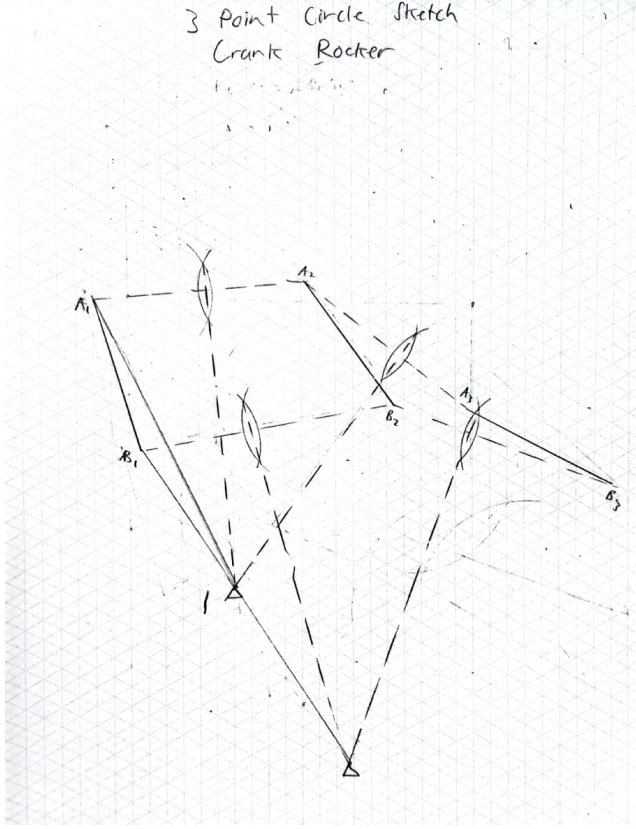


Fig. 2: Three-point circle construction sketches.

B. Optimization and Decision Rationale

Design selection prioritized: (1) ability to reach target point while remaining inside the frame, (2) minimal positional error, (3) manufacturability in 1/4" MDF.

Software optimization method: The MATLAB `fmincon` solver was used with constrained nonlinear minimization. The objective function was

$$f(\mathbf{x}) = \sum_{i=1}^N \sqrt{(x_{ee,i} - x_t)^2 + (y_{ee,i} - y_t)^2} \quad (3)$$

where $\mathbf{x}$ contains the link lengths and geometry parameters, $(x_{ee,i}, y_{ee,i})$ are the computed end-effector positions at input angles $\theta_1^{(i)}$, and $(x_t, y_t) = (152.4, 240)$ mm is the target. Constraints enforced frame boundaries and slit clearance.

The solver produced link values consistent with the initial set (within small adjustments) and verified the mechanism classification as a **crank-rocker** four-bar linkage capable of the required crank rotation range.

Input (servo) angles used for prototyping were:

$$\theta_{\text{start}} = 157^\circ \quad (\text{retracted}), \quad \theta_{\text{end}} = 5^\circ \quad (\text{extended}),$$

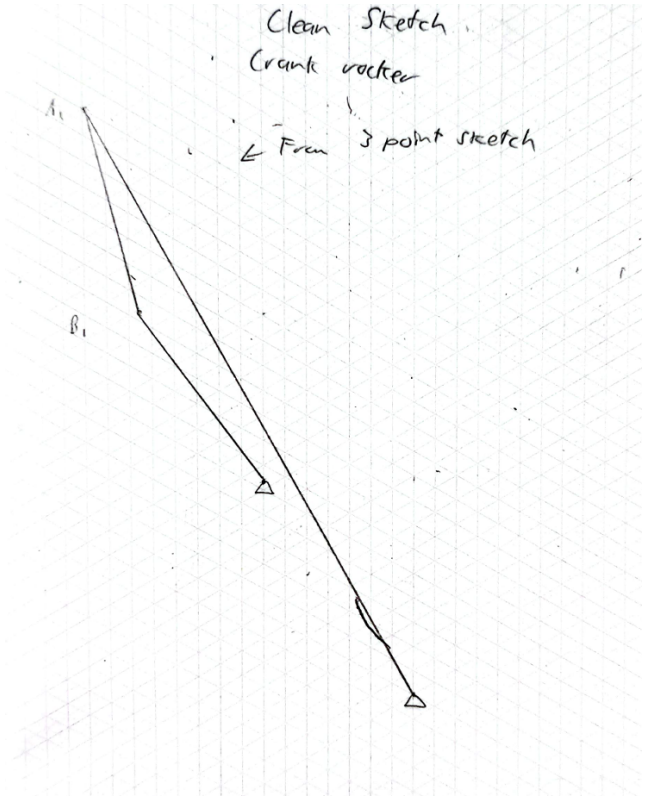


Fig. 3: Clean three-point circle sketch.

TABLE I: Final link geometry used for the prototype (mm).

Component	Length (mm)
Crank (L_2)	68
Coupler (L_3)	75
Follower (L_4)	85
Ground (L_1)	87
End effector reach (r_{ee})	249.3

consistent with the MATLAB model and Arduino servo mapping.

III. ANALYSIS

A. End-Effector Path and Motion

The positions of all links were calculated for each input crank angle θ_1 . This produced the path of the end effector from retracted to extended positions. The computed path was compared to the target point $(x_t, y_t) = (152.4 \text{ mm}, 240 \text{ mm})$. The positional difference was ≤ 0.05 mm across the sweep.

Figure 4 shows the initial four-bar geometry before optimization, while Figure 5 demonstrates the optimized trajectory. The optimization refined the link parameters to achieve ≤ 0.05 mm positional error at the target, improving upon the initial design's estimated 5 mm error. The optimization also increased mechanical advantage, decreasing torque.

B. Free Body Diagrams

Figure 6 shows the free body diagrams for each link, illustrating the forces and interactions considered in torque

Sketch → From Animation
Crank Rocker

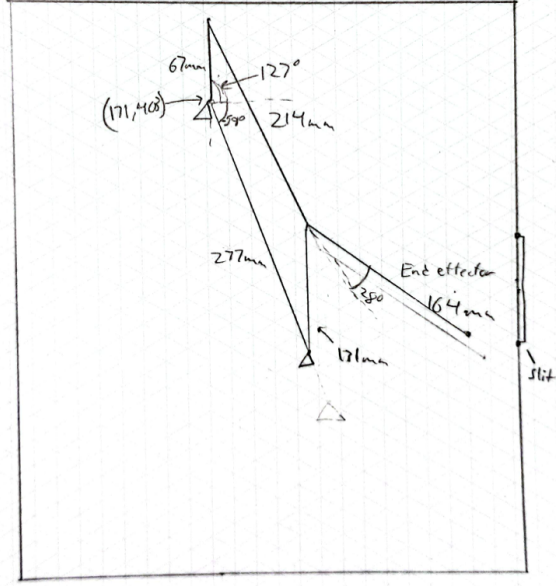


Fig. 4: Four-bar mechanism sketch prior to MATLAB optimization.

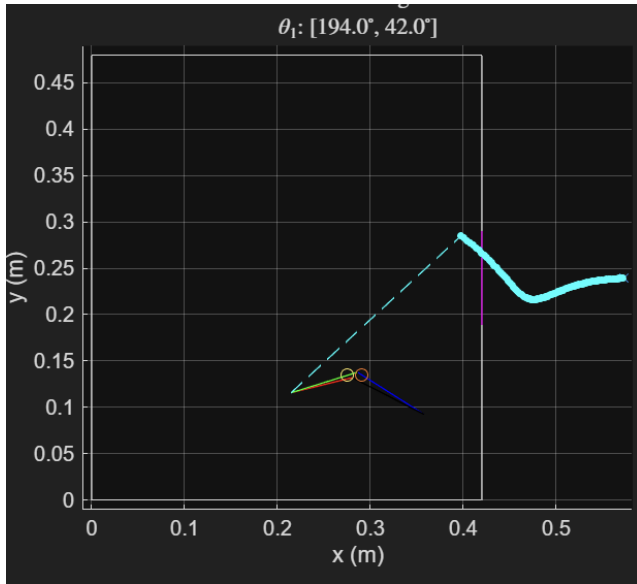


Fig. 5: End-effector trajectory from MATLAB optimization. Target is indicated by cross.

analysis.

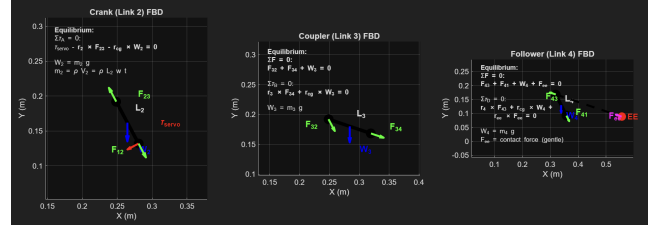


Fig. 6: Free body diagrams showing forces on each link.

C. Assumed Loads and Link Properties

- Links approximated as rigid bodies with concentrated masses at centroids (MDF thickness 1/4", typical link widths).
- End-effector contact force assumed small but bounded for worst-case torque.
- Joint friction and tolerances add minor torque margins.

D. Servo Torque Analysis

Torque at the servo:

$$\tau(\theta_1) = \mathbf{r}_c(\theta_1) \times \mathbf{F}_{eq}(\theta_1) \quad (4)$$

HS-425BB torque ≈ 470 N-mm. Figure 7 shows the required servo torque vs input crank angle, with the peak below the servo rating.

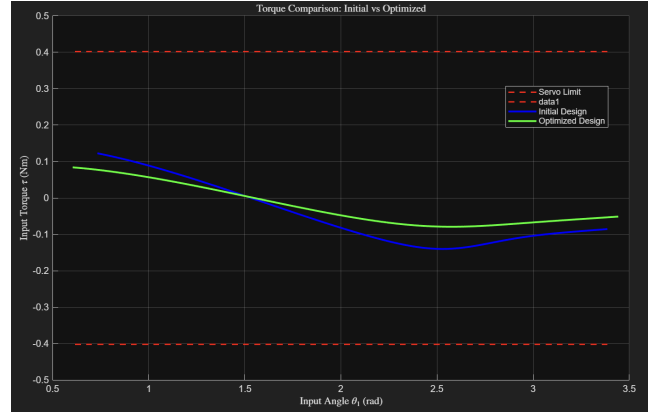


Fig. 7: Required servo torque vs input crank angle.

E. Trajectory Verification vs Constraints

- Path stays within frame.
- Intersects target.
- Exits slit correctly.

IV. EXPERIMENTAL RESULTS

A. Prototype Fabrication

1/4" MDF laser-cut with shoulder screws used. The prototype was mounted to a foam-core frame with simulations indicate comparable performance.

B. Control, Sensing, Reliability

An Arduino TSOP38238 IR receiver was used, and IDE code included 500 ms debounce and protocol validation applied.

C. Performance Results

- Target reached ± 2 mm.
- MATLAB predicted ≤ 0.05 mm error.
- 10 cycles completed reliably.
- Servo smooth, no stall.
- Gentle end-effector contact.

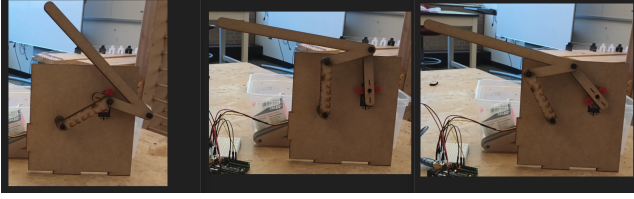


Fig. 8: Experimental sequence: (a) retracted, (b) extending, (c) at charging position.

D. Analysis of What Went Right/Wrong

What went right:

- Smooth motion via 1° stepping eliminated jarring.
- IR filtering eliminated false triggers.
- Mechanism stayed within frame constraints.
- Servo detach strategy eliminated idle noise.

What went wrong and corrections:

- Initial servo jump corrected using `moveTo()`.
- IR noise triggers corrected by protocol validation and debounce.
- 2 mm error due to MDF compliance and clearances; could improve with stiffer material.

V. DISCUSSION

A. Verification Against Functional Requirements

TABLE II: Functional requirement verification checklist

Requirement	Notes	Pass
Frame retraction	Mechanism fully inside frame	✓
Slit constraint	Motion exits via ≤ 101.6 mm slit	✓
Target reach	152.4 mm at 240 mm height	✓ (± 2 mm)
IR trigger	TSOP38238 detects reliably	✓
Smooth motion	<code>moveTo()</code> $1^\circ / 15$ ms	✓
Hold times	≥ 3 s	✓
Repeatability	10+ cycles	✓
Gentle contact	No deflection	✓

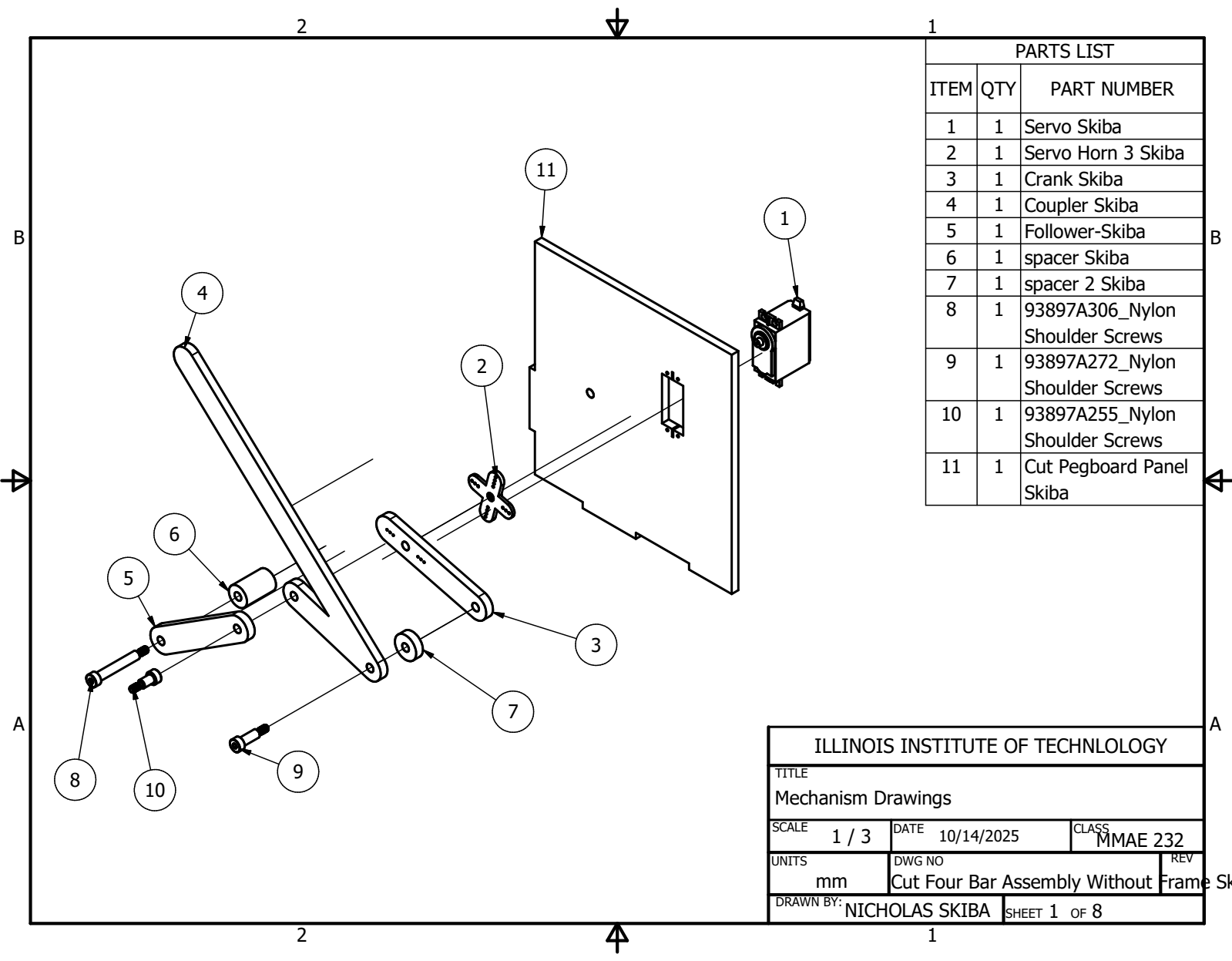
B. Arduino Control Summary

The mechanism is controlled by an Arduino Uno, which receives signals from a TSOP38238 IR receiver and smoothly drives the HS-425BB servo using incremental motion with `moveTo()`, while managing debounce and LED indicators.

VI. CONCLUSIONS

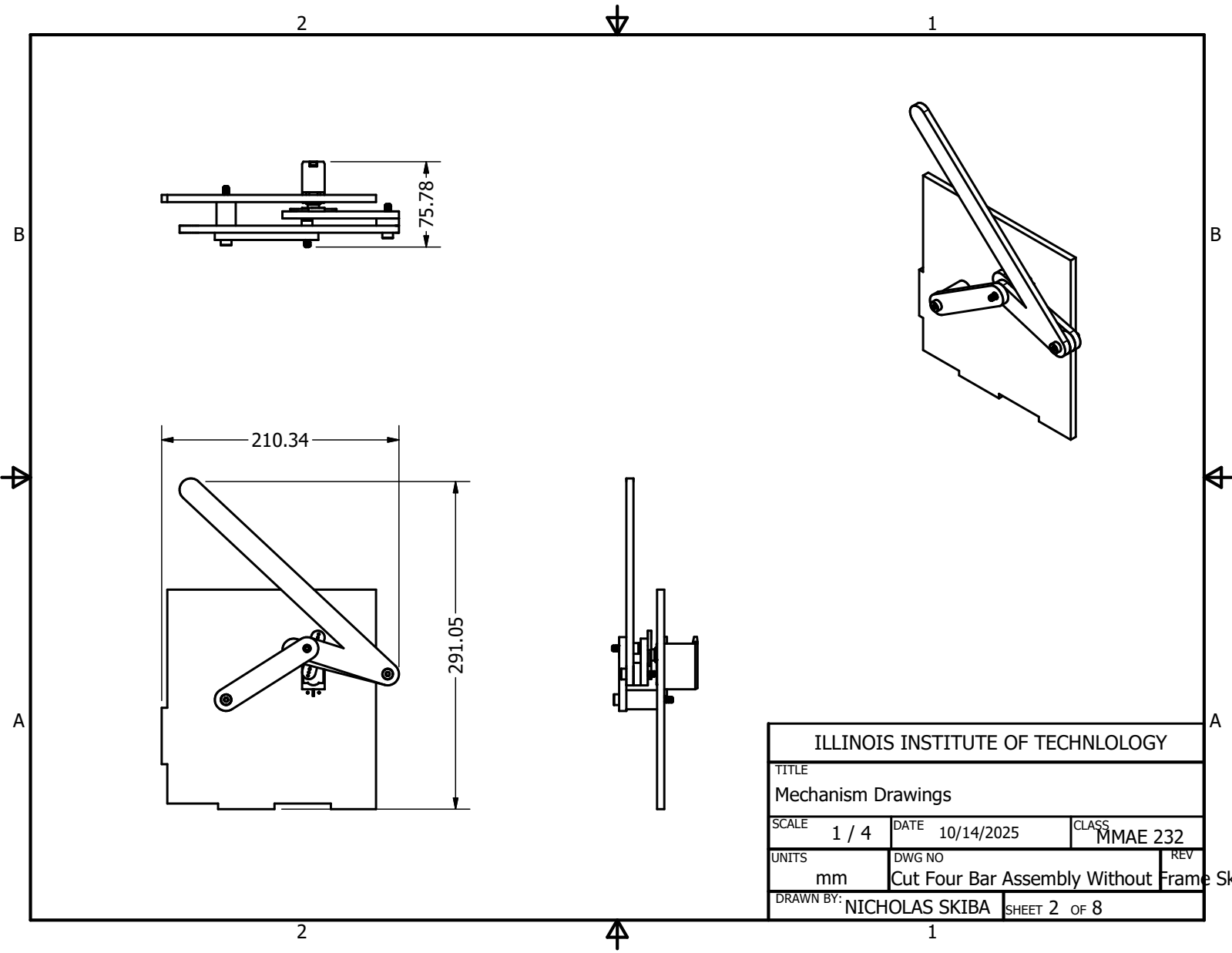
A compact four-bar mechanism was synthesized, optimized, prototyped. MATLAB was used to get an optimized ≤ 0.05 mm theoretical offset from center, and a physical ± 2 mm offset. The IR-triggered control was reliable and the torque margin sufficient. The project demonstrates practical low-complexity autonomous EV charging.

APPENDIX A ENGINEERING DRAWINGS

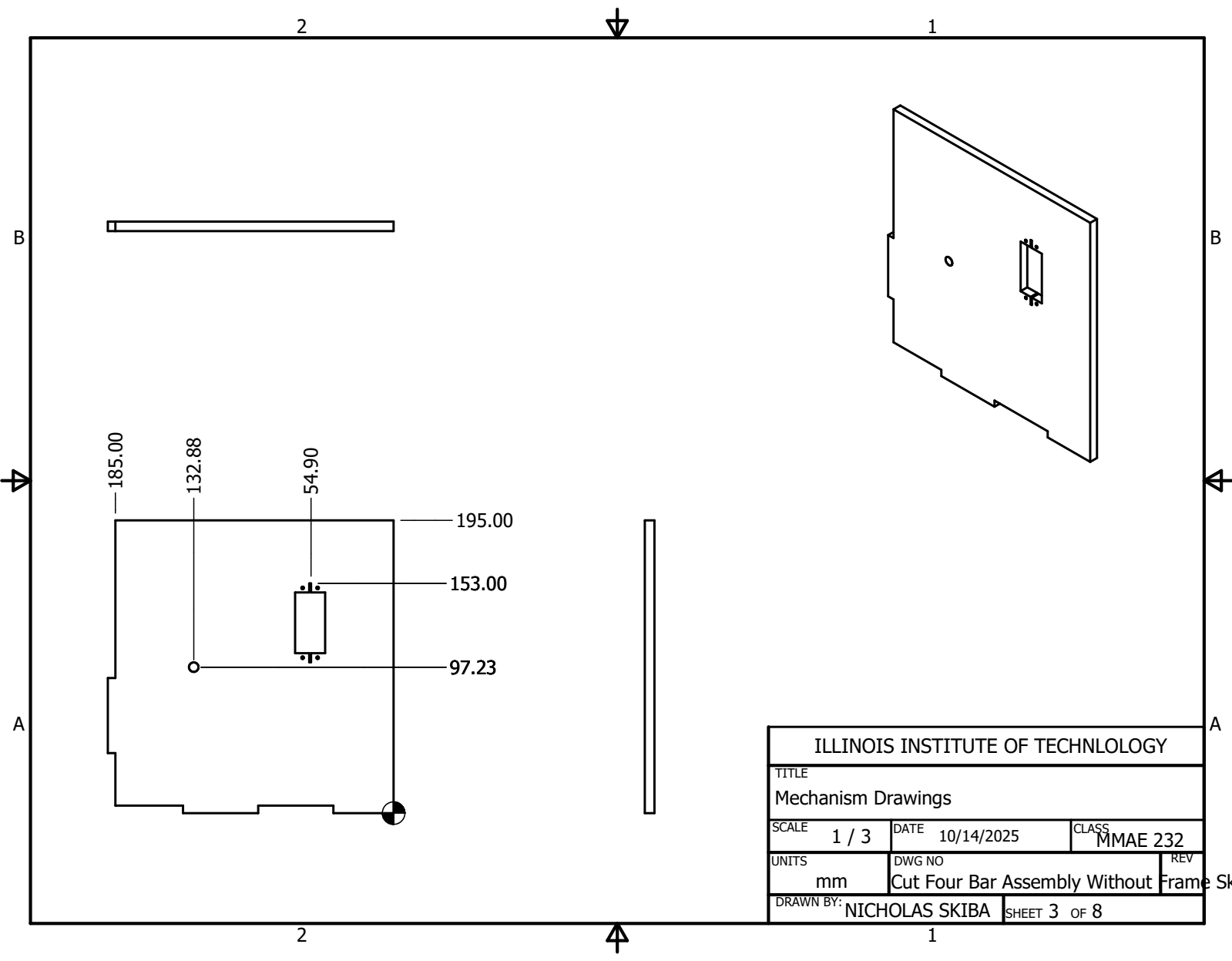


PARTS LIST		
ITEM	QTY	PART NUMBER
1	1	Servo Skiba
2	1	Servo Horn 3 Skiba
3	1	Crank Skiba
4	1	Coupler Skiba
5	1	Follower-Skiba
6	1	spacer Skiba
7	1	spacer 2 Skiba
8	1	93897A306_Nylon Shoulder Screws
9	1	93897A272_Nylon Shoulder Screws
10	1	93897A255_Nylon Shoulder Screws
11	1	Cut Pegboard Panel Skiba

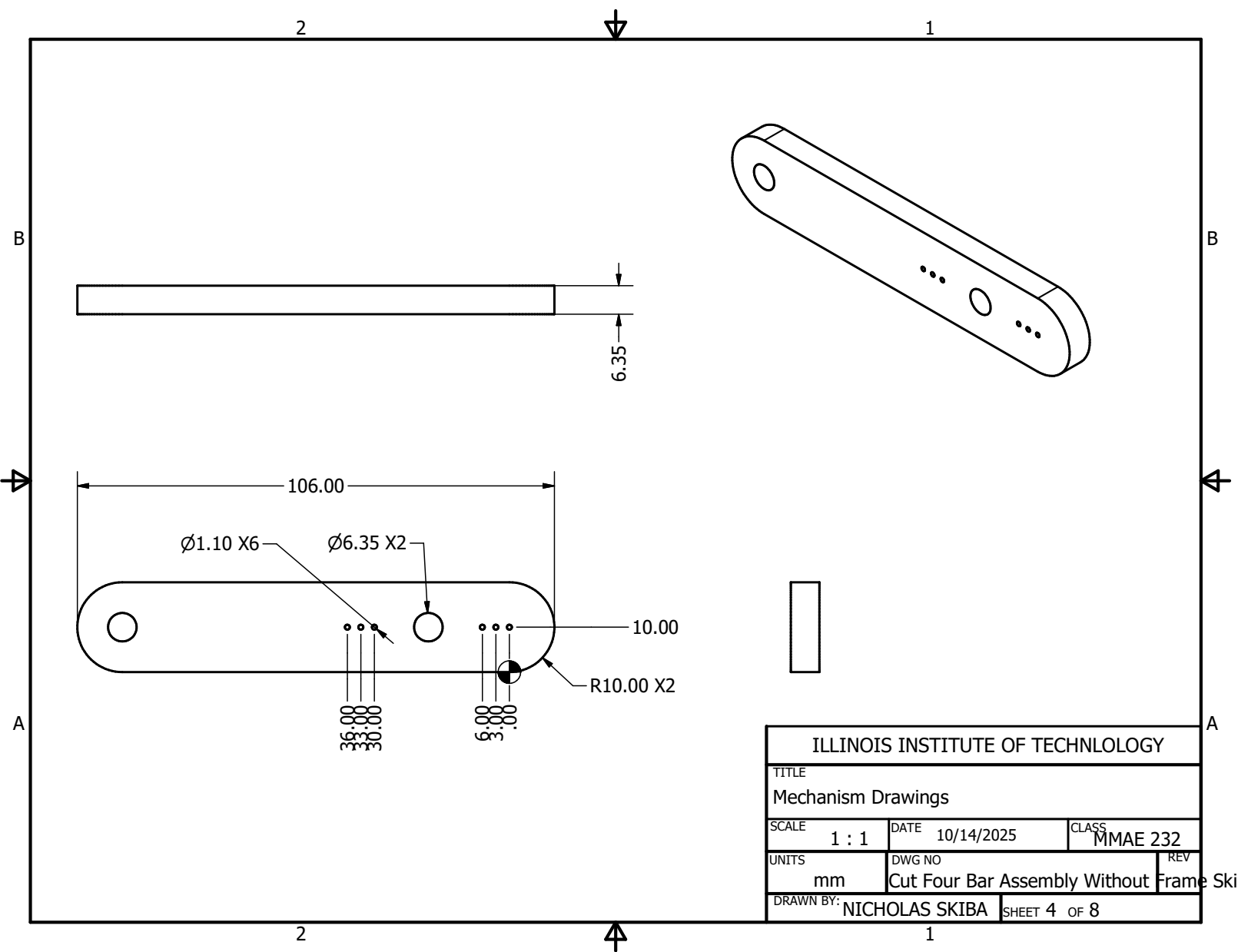
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SCALE 1 / 3	DATE 10/14/2025	CLASS MMAE 232	
UNITS mm	DWG NO Cut Four Bar Assembly Without Frame Skib	REV	
DRAWN BY: NICHOLAS SKIBA		SHEET 1 OF 8	

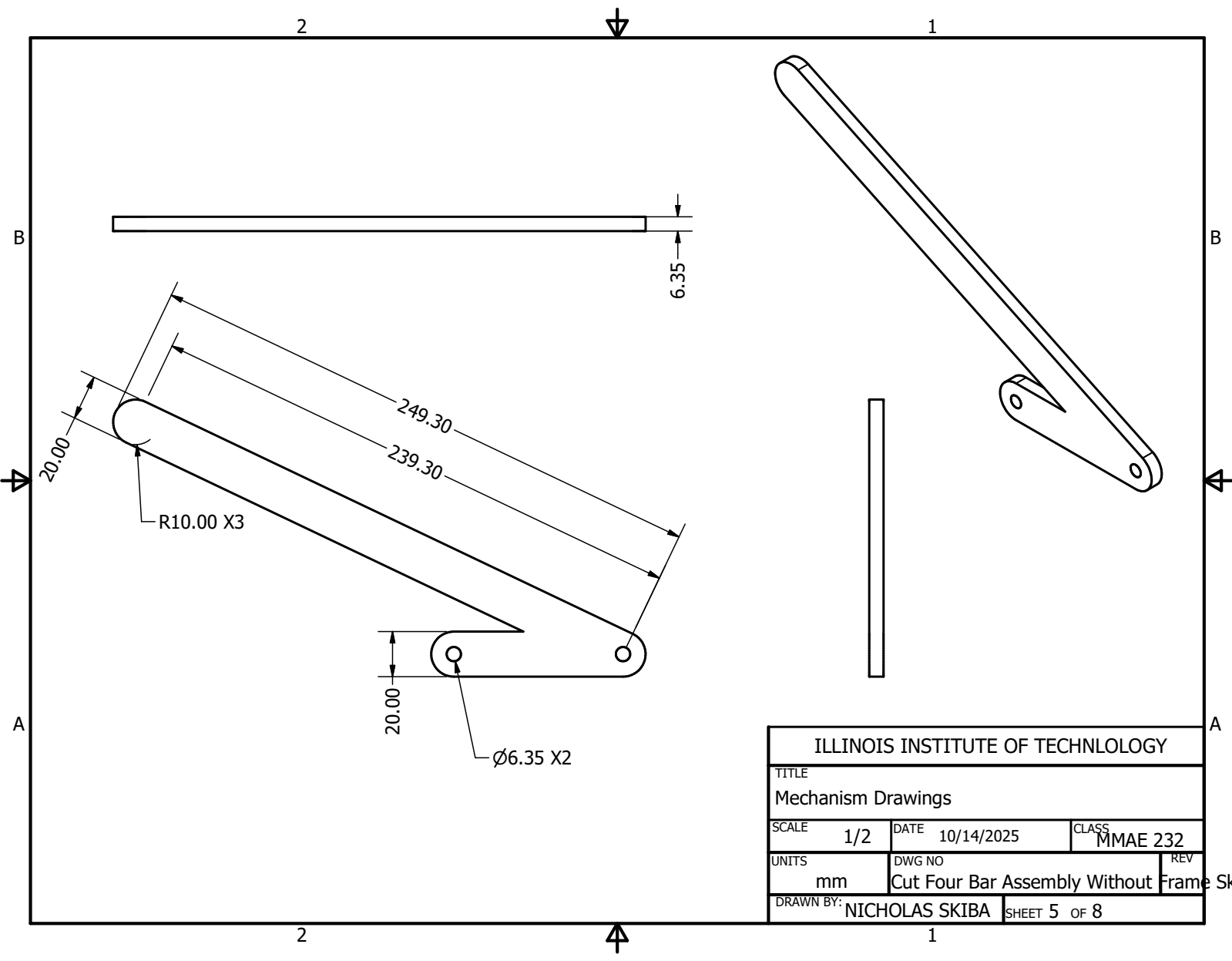


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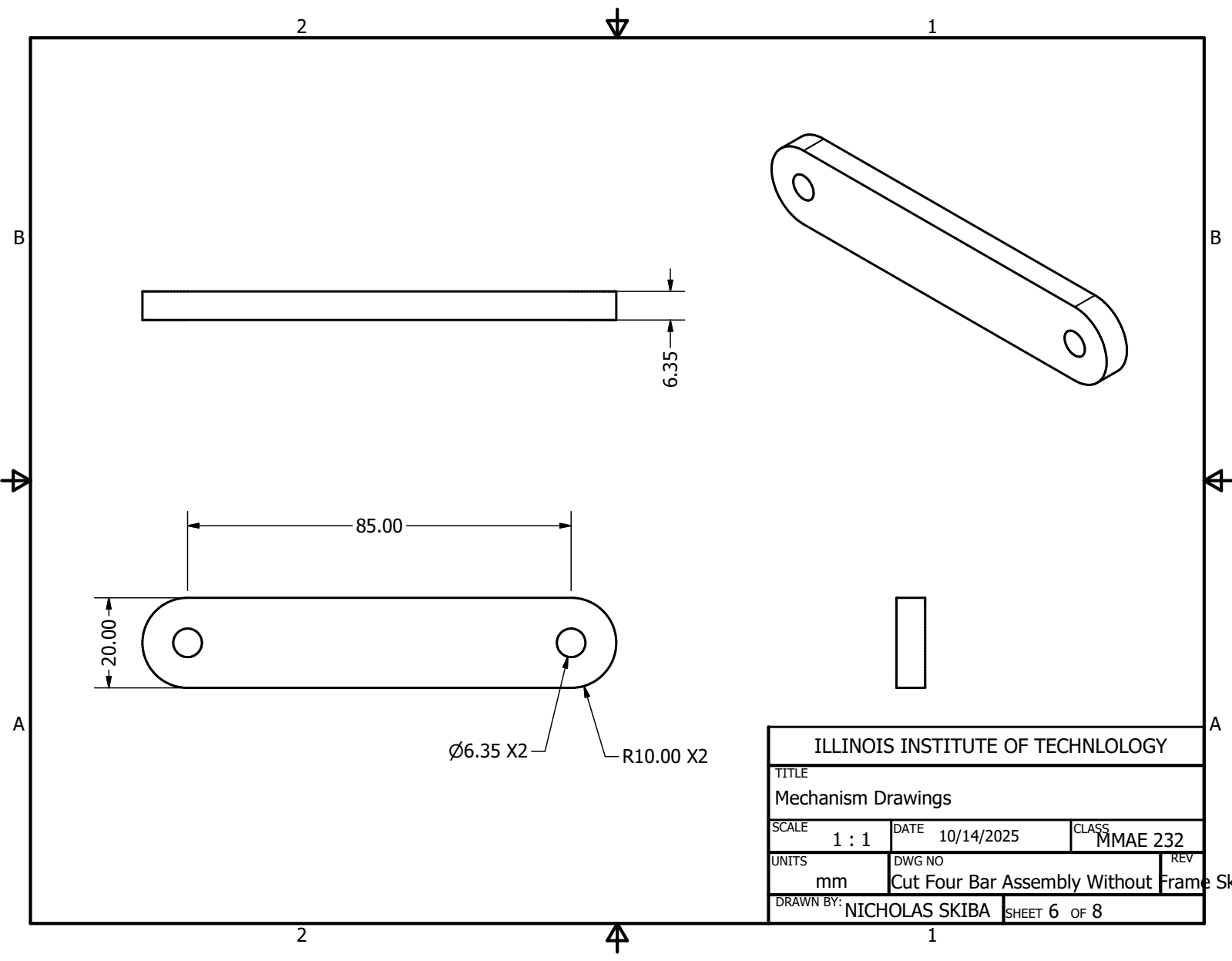


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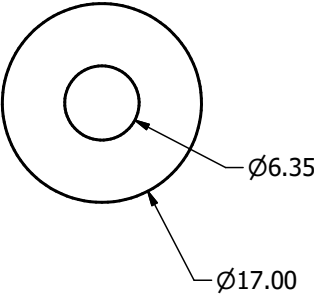
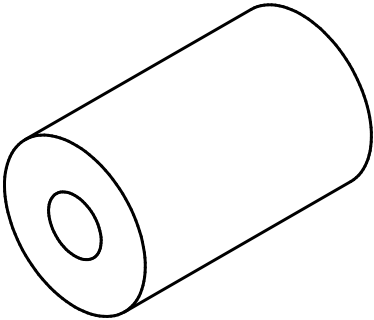
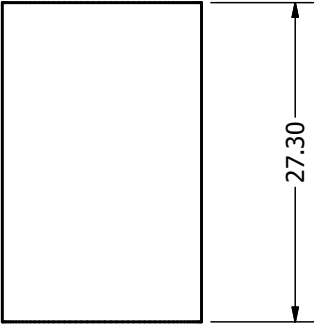




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TITLE Mechanism Drawings			
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UNITS mm	DWG NO Cut Four Bar Assembly Without Frame Sk	REV	
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TITLE			
Mechanism Drawings			
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		CLASS	MMAE 232
UNITS	mm	DWG NO	Cut Four Bar Assembly Without Frame Skib
DRAWN BY:	NICHOLAS SKIBA	SHEET	7 OF 8

